

C650

Geology I: Physical

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student on a field trip to a quarry that has operated since 1950 finds a gray mineral that scratches glass (about 5.5) but is scratched by quartz (7). The quarry has a gift shop. What is the mineral's hardness most likely to be?

- A. About 3, like calcite
- B. Exactly 7, since it came from a quarry
- C. About 6, such as orthoclase feldspar
- D. Above 7, like topaz

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Two gold-colored metallic samples may be gold or pyrite. Which test best distinguishes them?

- A. Luster
- B. Color
- C. None; only a professional lab can tell them apart
- D. Streak

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A sandstone layer beneath a mountain range that attracts thousands of climbers is buried deeply and subjected to high heat and pressure without melting. The range is in a national park. What does the sandstone become?

- A. A metamorphic rock (quartzite)
- B. Magma, since deep burial always melts rock
- C. It stays sedimentary
- D. An igneous rock

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A countertop maker who ships to five states uses granite with large, visible crystals. The stone is polished to a high shine. What does the crystal size show about how the granite formed?

- A. Crystal size depends only on mineral color.
- B. It cooled slowly deep underground.
- C. It cooled quickly at the surface.
- D. It formed from compacted sediment.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Obsidian is a glassy igneous rock with no visible crystals. How did it cool?

- A. Slowly, then suddenly
- B. Very quickly at or near the surface
- C. Slowly, deep underground
- D. Through metamorphism

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

The Andes, where a mining company employs 8,000 workers, have a deep ocean trench offshore and a chain of volcanoes inland. The company exports copper. What kind of plate boundary formed them?

- A. Continent-continent collision
- B. Transform
- C. Divergent
- D. Convergent, with an oceanic plate subducting beneath a continent

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A boundary has frequent earthquakes and plates sliding past each other horizontally, with no volcanoes. What type is it?

- A. Convergent
- B. A hot spot
- C. Transform
- D. Divergent

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a warm, rainy region known for its coffee farms, granite bedrock slowly turns to soft clay as its feldspar changes. The region gets 250 cm of rain a year. What process is at work?

- A. Mechanical weathering by frost wedging
- B. Wind abrasion
- C. Chemical weathering
- D. None; granite is too hard to weather

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A river that powers an old mill slows as it enters a lake. It carries gravel, sand, silt, and clay. The mill is now a museum. Which sediment settles first, closest to the river mouth?

- A. Gravel
- B. Dissolved minerals
- C. Clay
- D. All sizes settle together

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A well driller whose family has run the business for three generations finds a thick clay layer that is 50% pore space. The drilling rig is new. Why will a well in this clay yield very little water?

- A. High porosity always means high permeability.
- B. Clay stores much water but its tiny pores let water flow through very poorly.
- C. Clay has very low porosity.
- D. Clay transmits water faster than sand.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which material would make the most productive aquifer?

- A. Well-sorted gravel
- B. Unfractured granite
- C. Clay
- D. Shale

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A geologist mapping a valley where a wine region has 40 vineyards finds a fault where the hanging wall moved down relative to the footwall as the crust was stretched. The vineyards export to Europe. What type of fault is this?

- A. Reverse fault
- B. Strike-slip fault
- C. Thrust fault, since all faults near mountains are thrust faults
- D. Normal fault

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A student on a field trip to a quarry ~~that has operated since 1950~~ finds a gray mineral that scratches glass (about 5.5) but is scratched by quartz (7). ~~The quarry has a gift shop.~~ What is the mineral's hardness most likely to be?

Target word: hardness range **Decides:** Between glass and quartz.

A	Dropped element	Ignores that it scratched glass.
B	Camouflage	The quarry detail does not decide hardness; quartz scratched it.
C	Correct	It is harder than glass and softer than quartz.
D	Inverted operation	A mineral harder than quartz would scratch quartz, not be scratched by it.

2. Answer D Concept 1 • transfer

Two gold-colored metallic samples may be gold or pyrite. Which test best distinguishes them?

Target word: best identifying test **Decides:** Streak differs: gold-yellow vs. greenish-black.

A	Adjacent category	Both have metallic luster.
B	Confusion set	Both look gold, so color cannot separate them.
C	Absolute language	A simple streak test separates them.
D	Correct	Gold leaves a yellow streak; pyrite leaves a greenish-black streak.

3. Answer A Concept 2

A sandstone layer beneath a mountain range ~~that attracts thousands of climbers~~ is buried deeply and subjected to high heat and pressure without melting. ~~The range is in a national park.~~ What does the sandstone become?

Target word: rock type after heat and pressure **Decides:** No melting means metamorphism.

A	Correct	Heat and pressure without melting recrystallize it.
B	Absolute language	Rock can be transformed well below its melting point.
C	Dropped element	Ignores the heat and pressure.
D	Adjacent category	Igneous rock requires melting and cooling.

4. Answer B Concept 3

A countertop maker ~~who ships to five states~~ uses granite with large, visible crystals. ~~The stone is polished to a high shine.~~ What does the crystal size show about how the granite formed?

Target word: cooling history **Decides:** Large crystals mean slow cooling.

A	Adjacent category	Color does not control crystal size.
B	Correct	Slow cooling gives crystals time to grow.
C	Inverted operation	Fast cooling makes small crystals.
D	Confusion set	That describes sedimentary rock.

5. Answer B Concept 3 • transfer

Obsidian is a glassy igneous rock with no visible crystals. How did it cool?

Target word: reverse: cooling from texture **Decides:** Glass means no time to crystallize.

A	Adjacent category	That two-stage history produces porphyritic texture.
B	Correct	Rapid cooling prevents crystal growth.
C	Inverted operation	That gives large crystals.
D	Confusion set	Obsidian forms from cooling lava, not metamorphism.

6. Answer D Concept 4

The Andes, ~~where a mining company employs 8,000 workers~~, have a deep ocean trench offshore and a chain of volcanoes inland. ~~The company exports copper~~. What kind of plate boundary formed them?

Target word: boundary type **Decides:** Trench plus volcanoes.

A	Adjacent category	That builds high mountains, like the Himalayas, with few volcanoes.
B	Confusion set	Sideways sliding makes earthquakes but not trenches and volcanic arcs.
C	Inverted operation	Divergent boundaries pull apart and form ridges, not trenches.
D	Correct	Subduction makes the trench and feeds the volcanoes.

7. Answer C Concept 4 • transfer

A boundary has frequent earthquakes and plates sliding past each other horizontally, with no volcanoes. What type is it?

Target word: reverse: boundary from features **Decides:** Sideways motion.

A	Confusion set	Convergent boundaries often build volcanoes and mountains.
B	Adjacent category	Hot spots are not boundaries and produce volcanoes.
C	Correct	Plates slide past each other.
D	Inverted operation	Plates move apart and often erupt.

8. Answer C Concept 5

In a warm, rainy region ~~known for its coffee farms~~, granite bedrock slowly turns to soft clay as its feldspar changes. The region gets 250 cm of rain a year. What process is at work?

Target word: type of weathering **Decides:** A mineral changes to a new mineral.

A	Confusion set	Frost wedging breaks rock without changing minerals and needs freezing.
B	Adjacent category	Physical wearing, not a mineral change.
C	Correct	Feldspar reacting with water to form clay changes the minerals.
D	Absolute language	All rock weathers, especially in warm, wet climates.

9. Answer A Concept 6

A river ~~that powers an old mill~~ slows as it enters a lake. It carries gravel, sand, silt, and clay. ~~The mill is now a museum.~~ Which sediment settles first, closest to the river mouth?

Target word: first deposit **Decides:** Slowing water drops the largest particles first.

A	Correct	The heaviest particles need the most energy to carry.
B	Adjacent category	Dissolved load stays in solution; it does not settle as the water slows.
C	Inverted operation	Clay is the finest and settles last, farthest out.
D	Absolute language	Deposition sorts sediment by size.

10. Answer B Concept 7

A well driller ~~whose family has run the business for three generations~~ finds a thick clay layer that is 50% pore space. ~~The drilling rig is new.~~ Why will a well in this clay yield very little water?

Target word: why clay yields little **Decides:** High porosity, low permeability.

A	Absolute language	Clay is the classic exception.
B	Correct	Permeability, not porosity, controls well yield.
C	Confusion set	The stem says 50% pore space; its problem is permeability.
D	Inverted operation	Sand transmits water much faster.

11. Answer A Concept 7 · transfer

Which material would make the most productive aquifer?

Target word: reverse: pick high permeability **Decides:** Large, connected pores.

A	Correct	Large connected pores give high permeability.
B	Adjacent category	Very little connected pore space.
C	Confusion set	Porous but nearly impermeable.
D	Dropped element	Fine grains and tight layers block flow.

12. Answer D Concept 8

A geologist mapping a valley ~~where a wine region has 40 vineyards~~ finds a fault where the hanging wall moved down relative to the footwall as the crust was stretched. ~~The vineyards export to Europe.~~ What type of fault is this?

Target word: fault type **Decides:** Tension with hanging wall down.

A	Inverted operation	Compression pushes the hanging wall up.
B	Confusion set	Horizontal sliding from shear stress.
C	Absolute language	Fault type depends on stress, not setting.
D	Correct	Tension drops the hanging wall.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Mineral identification tests	Hardness (Mohs scale, by scratching) and streak are more reliable than color; a mineral scratches anything softer than itself.	Identifying a mineral by color alone or reversing who scratches whom.	A scratch test result or two minerals that look alike.
2	The rock cycle	Heat and pressure without melting make metamorphic rock; melting and cooling make igneous rock; weathering, deposition, and lithification make sedimentary rock.	Thinking deep burial always melts rock into magma.	A rock buried deeply or exposed at the surface.
3	Cooling rate and igneous texture	Slow cooling underground grows large crystals (intrusive, coarse); fast cooling at the surface makes small crystals or glass (extrusive).	Linking large crystals to fast cooling.	Crystal size in granite, basalt, or obsidian.
4	Plate boundaries and their features	Oceanic-continental convergence forms trenches and volcanic mountains; divergence forms ridges and rifts; transform boundaries slide past with earthquakes but little volcanism.	Assigning volcanoes to transform boundaries or trenches to divergent ones.	A named landform with trench, volcano, rift, or sideways motion.
5	Mechanical vs. chemical weathering	Mechanical weathering breaks rock into smaller pieces without changing its minerals; chemical weathering changes minerals (for example, feldspar to clay) and is fastest in warm, wet climates.	Assuming hard rock does not weather, or mixing up the two types.	Climate details and a changed or unchanged mineral.
6	Stream velocity and deposition	As a stream slows, it drops its largest particles first and the finest last, sorting sediment by size.	Thinking fine clay settles first or all sizes drop together.	A river entering a lake or the sea.
7	Porosity vs. permeability	Porosity is how much pore space a material has; permeability is how easily water flows through it. Clay can be very porous but nearly impermeable.	Assuming high porosity always means high permeability.	Clay, sand, or gravel and a well or aquifer.
8	Fault types and stress	Tension makes normal faults (hanging wall moves down); compression makes reverse or thrust faults (hanging wall moves up); shear makes strike-slip faults.	Swapping normal and reverse faults.	Hanging wall direction plus stretching or squeezing.

Course scope (public description): Geology I: Physical provides undergraduate students seeking initial licensure or endorsement in secondary science education with an introduction to minerals and rocks, the physical features of the Earth, and the internal and surface processes that shape those features. This course has no

prerequisites.